

Knowing When Not to Answer: Evaluating Confidence and NLI-Based Verification for Selective Question Answering

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Abstract

Selective question answering allows a system to abstain from predictions that are likely to be incorrect rather than answering every input. This work investigates whether semantic verification provides a better basis for selective abstention than model confidence alone. We evaluate a pretrained extractive QA model on SQuAD v2 using a controlled protocol in which all methods rank the same forced-answer candidates. The comparison includes temperature-scaled model confidence, a question-aware verification pipeline that converts question-answer pairs into declarative claims before natural language inference, an answer-support NLI verifier, and two fixed confidence-verification fusions. Selective performance is measured primarily by area under the risk-coverage curve (AURC), where lower values indicate better ranking. On the final held-out set of 3,000 examples, confidence achieves the best global AURC of 0.292379, compared with 0.309120 for confidence plus answer-support verification and 0.339417 for confidence plus question-aware verification; both verifier-only rankings perform worse. Paired bootstrap analysis with 5,000 resamples supports the stability of the confidence-only advantage on this held-out sample. Error and pairwise ranking analyses show that the verification signals are diagnostically informative but introduce more harmful than beneficial reordering when fused with confidence. These results demonstrate that additional semantic verification does not necessarily improve selective QA ranking over a strong confidence baseline, even when the verifier itself provides meaningful support information.

1 Introduction

Question-answering systems are often evaluated by how frequently they produce correct answers, but reliability also depends on recognizing when an answer should not be given. In realistic settings, a model may encounter questions for which the available context does not support a reliable answer, yet still produce a plausible-looking prediction. This motivates selective question answering, where a system is allowed to abstain on uncertain predictions rather than answering every input. The goal is therefore not only to maximize accuracy, but to identify unreliable predictions well enough to trade coverage for lower error rates [2, 4]. This problem is especially relevant to SQuAD v2, which explicitly includes unanswerable questions and requires systems to determine when no supported answer exists in the provided context [7].

A natural way to support abstention is to use the model’s own confidence as a ranking signal: predictions with higher confidence are answered first, while lower-confidence predictions are rejected as coverage decreases. However, neural-network confidence scores are not necessarily well calibrated, meaning that a model can assign high confidence to incorrect predictions [3]. Post-hoc calibration methods such as temperature scaling can improve the relationship between predicted confidence and empirical correctness without retraining the underlying model. For selective question answering,

calibrated confidence therefore provides a simple and strong baseline for deciding which predictions should be retained and which should be abstained from.

Confidence, however, reflects the model’s internal preference for a prediction rather than directly testing whether the proposed answer is semantically supported by the available context. This raises the possibility that an additional verification signal could improve abstention decisions. Natural language inference (NLI) provides a natural mechanism for such verification by estimating whether a hypothesis is entailed by, contradicted by, or neutral with respect to a premise [9]. In question answering, this suggests a complementary strategy: instead of relying only on how confident the QA model is, the system can also examine whether the context supports the meaning of the proposed answer. The central question of this work is therefore whether such semantic and answer-support verification signals can identify unreliable QA predictions better than calibrated confidence alone, thereby improving selective abstention.

To investigate this question, we conduct a controlled selective-QA study on SQuAD v2 using the pretrained `deepset/roberta-base-squad2` extractive QA model. We compare calibrated model confidence with two NLI-based verification signals: a question-aware semantic verifier that converts a question–answer pair into a declarative claim before testing whether the context entails it, and an answer-support verifier that directly assesses whether the proposed answer is supported by the context. These signals are evaluated individually and in fixed confidence–verification combinations, giving five ranking methods in total. All methods rank the same underlying forced-answer QA candidates, allowing differences in selective performance to be attributed to the ranking signal rather than to changes in the QA predictions themselves. Performance is evaluated primarily using the area under the risk–coverage curve (AURC), where lower values indicate a better ability to retain correct predictions while abstaining on unreliable ones.

On the final held-out evaluation of 3,000 examples, calibrated confidence provides the strongest overall selective ranking, achieving an AURC of 0.292379. Adding the answer-support verifier increases AURC to 0.309120, while combining confidence with the question-aware semantic verifier increases it to 0.339417; both verifier-only rankings perform worse still. Paired bootstrap analysis with 5,000 resamples yields positive AURC differences relative to confidence-only ranking for all four alternatives on the final held-out set, indicating that the observed global ordering is stable under resampling. At the same time, smaller subsets and individual coverage levels show local reversals, demonstrating that verification signals can still be informative without improving the complete selective ranking. The main contribution of this work is therefore a controlled evaluation showing that additional NLI-based verification does not necessarily lead to better abstention decisions than a strong calibrated-confidence baseline, and an analysis of why semantically meaningful verification signals can nevertheless degrade global risk–coverage performance.

2 Related Work

Selective prediction and abstention. Selective prediction studies models that are allowed to reject uncertain inputs rather than producing a prediction for every example. Geifman and El-Yaniv formalize this setting through the trade-off between coverage and risk, showing how a trained neural network can selectively retain predictions while controlling error [2]. This perspective is directly relevant to question answering, where abstention can be preferable to returning an unsupported answer. Kamath et al. extend selective prediction explicitly to question answering under domain shift and show that abstention policies based only on raw model probabilities can be unreliable when models become overconfident on unfamiliar inputs [4]. Our work adopts the same general selective-prediction perspective, but focuses on a controlled in-domain held-out comparison of calibrated

confidence and NLI-based verification signals rather than domain-shift calibration.

Confidence calibration. Confidence-based abstention depends on the quality of the score used to rank predictions. Neural networks can be poorly calibrated, so a high predicted confidence does not necessarily correspond to a high empirical probability of correctness [3]. Guo et al. study several post-hoc calibration methods and identify temperature scaling as a simple and effective approach for improving probabilistic calibration without retraining the underlying model. In our experiments, temperature scaling is fitted only on the calibration partition and then frozen before held-out evaluation. Calibrated confidence serves as the principal baseline against which the additional verification signals are compared.

NLI-based answer verification. Natural language inference has also been used as a mechanism for verifying whether a proposed answer is supported by textual evidence. Chen et al. reformulate question-answering instances as premise-hypothesis pairs and use NLI models to evaluate whether the document context supports a proposed answer [1]. They show that NLI-based verification can improve confidence estimation across domains in a selective-QA setting, motivating the use of semantic support as an additional signal beyond the QA model’s own confidence. Related work on textual fact verification, such as FEVER, similarly frames verification as deciding whether textual evidence supports, refutes, or provides insufficient information for a claim [8].

Relation to our study. Our study follows this general motivation but asks a narrower empirical question. We do not train a new NLI verifier or evaluate evidence retrieval; instead, the context is already given by the extractive-QA task and pretrained verification components are applied post hoc to the same fixed QA candidates. We then compare these verification signals directly against calibrated QA confidence under one controlled held-out protocol. The primary objective is not verifier classification accuracy itself, but whether the resulting score produces a better selective ordering as measured by risk-coverage performance. This distinction is important because a verifier may assign semantically meaningful support labels while still reordering correct and incorrect QA candidates in a way that worsens global AURC.

3 Methodology

3.1 Selective Question Answering

Let a question-answering model produce a candidate answer for each input example together with a scalar ranking score. Selective question answering uses this score to determine which predictions should be answered and which should be abstained from. For a given coverage level, only the highest-ranked fraction of predictions is retained, while the remaining predictions are rejected.

In our final evaluation, all compared methods operate on the same forced-answer QA candidates. Candidate correctness is therefore fixed before ranking: for answerable examples, a prediction is correct when its normalized text exactly matches at least one reference answer; for unanswerable examples, the forced-answer candidate is treated as incorrect. The compared methods differ only in the score used to order these fixed candidates. This controlled design isolates the effect of the ranking signal from changes in the underlying QA predictions.

3.2 Calibrated Confidence

The QA backbone is the pretrained `deepset/roberta-base-squad2` model, based on the RoBERTa architecture [5]. For each forced-answer candidate, the system derives an answer-versus-null confidence signal from the model outputs. Because raw neural confidence can be poorly calibrated, we apply temperature scaling as a post-hoc calibration step [3].

The temperature parameter is fitted exclusively on the calibration partition and is then frozen before application to held-out test predictions. The fitted value used in the final experiments is

$$T = 4.604539.$$

Temperature scaling changes the probability scale but, because the transformation is monotonic for positive T , it does not alter the ordering induced by confidence alone. Calibrated confidence is therefore used as the principal baseline ranking signal throughout the final selective-QA evaluation.

3.3 Question-Aware Semantic Verification

The first verification signal evaluates whether the meaning of a proposed answer is supported by the supplied context. Because an isolated answer string is often not a complete proposition, we first combine the question and predicted answer into a declarative claim using the pretrained `domenicrosati/QA2D-t5-base` model, which is based on the T5 architecture [6].

Let q_i denote the question, a_i the forced-answer candidate, and c_i the corresponding context. The QA-to-declarative transformation produces a claim

$$h_i = \text{QA2D}(q_i, a_i).$$

Before NLI scoring, the generated claim is checked using deterministic structural validation rules. These checks are intended to reject malformed transformations, including cases where important answer content is not preserved. If the generated claim is invalid, its semantic verification score is set to

$$s_i^{\text{QA}} = 0.$$

For valid claims, the context c_i is treated as the NLI premise and the generated claim h_i as the hypothesis. We use the pretrained `FacebookAI/roberta-large-mnli` model, based on the RoBERTa architecture [5], and define the question-aware semantic ranking score as the predicted entailment probability:

$$s_i^{\text{QA}} = P_{\text{NLI}}(\text{ENTAILMENT} \mid c_i, h_i).$$

A high score therefore indicates that the context strongly supports the declarative claim constructed from the proposed answer, whereas a low score indicates weak or contradictory semantic support. This score is used directly as one of the five selective-ranking signals; the NLI label itself does not alter the underlying QA prediction or its correctness.

3.4 Answer-Support NLI Verification

The second verification signal evaluates answer support directly, without a generative QA-to-declarative step. For each non-empty candidate answer, we construct a deterministic hypothesis using the template

$$h_i^{\text{AS}} = \text{“The answer to the question } q_i \text{ is } a_i\text{.”}$$

The original context c_i is used as the NLI premise and the constructed answer-support statement h_i^{AS} as the hypothesis. As in the question-aware verifier, we use the pretrained **FacebookAI/roberta-large-mnli** model. The verifier produces entailment, neutral, and contradiction probabilities.

Rather than using entailment probability alone, the raw answer-support score contrasts semantic support with contradiction:

$$s_i^{\text{raw}} = P_{\text{NLI}}(\text{ENTAILMENT} \mid c_i, h_i^{\text{AS}}) - P_{\text{NLI}}(\text{CONTRADICTION} \mid c_i, h_i^{\text{AS}}).$$

This score lies in the interval $[-1, 1]$. For compatibility with the other ranking signals, it is mapped linearly to $[0, 1]$:

$$s_i^{\text{AS}} = \frac{s_i^{\text{raw}} + 1}{2}.$$

The verifier also assigns diagnostic labels of **SUPPORTED**, **UNCERTAIN**, or **REJECTED**, but these categorical labels are not used as the primary ranking score in the final AURC comparison. Instead, the continuous normalized score s_i^{AS} determines the selective ordering.

Although the question-aware and answer-support verification paths construct their hypotheses differently, they both use the same **FacebookAI/roberta-large-mnli** backbone. We therefore treat them as distinct verification signals rather than statistically independent verifier models.

3.5 Ranking Methods and Score Fusion

Let $c_i \in [0, 1]$ denote the calibrated confidence score for candidate i , s_i^{QA} the question-aware semantic verification score, and s_i^{AS} the normalized answer-support verification score. We evaluate five selective-ranking methods over the same set of QA candidates.

The three single-signal rankings are

$$r_i^{\text{Conf}} = c_i,$$

$$r_i^{\text{QA}} = s_i^{\text{QA}},$$

and

$$r_i^{\text{AS}} = s_i^{\text{AS}}.$$

To test whether verification complements model confidence, we additionally use two fixed equal-weight geometric-mean fusions:

$$r_i^{\text{Conf+QA}} = \sqrt{c_i s_i^{\text{QA}}},$$

and

$$r_i^{\text{Conf+AS}} = \sqrt{c_i s_i^{\text{AS}}}.$$

The geometric mean requires both components to contribute: a low verification score can therefore substantially reduce the combined ranking score even when model confidence is high. These

equal-weight fusion rules are fixed before held-out evaluation and constitute the two combined methods reported in the primary comparison.

As an auxiliary analysis, we also examine a weighted geometric family on the calibration partition only:

$$r_i(\alpha) = c_i^\alpha v_i^{1-\alpha},$$

where v_i denotes either verification score and

$$\alpha \in \{0.00, 0.01, \dots, 1.00\}.$$

The value of α is selected by minimizing calibration-set AURC, without using held-out test labels. For both verification paths, this search selects

$$\alpha = 1.00,$$

which corresponds to the confidence-only endpoint. This auxiliary result is kept separate from the primary fixed equal-weight comparison and provides an additional calibration-only check on whether the verifier signals improve ranking.

3.6 Risk–Coverage Evaluation and Abstention

A selective ranking becomes an abstention policy by retaining only the highest-scoring predictions. Let N denote the number of QA candidates and let

$$\pi(1), \pi(2), \dots, \pi(N)$$

denote the candidates sorted from highest to lowest ranking score. Score ties are resolved using the original deterministic evaluation index, ensuring that every method produces a reproducible ordering.

Let $y_i \in \{0, 1\}$ indicate whether candidate i is correct under the fixed forced-answer correctness definition. If only the first k ranked predictions are answered, the resulting coverage is

$$C_k = \frac{k}{N},$$

while all candidates ranked below k are treated as abstained predictions.

The selective accuracy among the retained predictions is

$$A_k = \frac{1}{k} \sum_{j=1}^k y_{\pi(j)},$$

and selective risk is the corresponding error rate,

$$R_k = 1 - A_k = \frac{1}{k} \sum_{j=1}^k (1 - y_{\pi(j)}).$$

A useful ranking therefore places correct predictions near the beginning of the ordering and incorrect predictions near the end. As coverage decreases, low-ranked candidates are abstained from first. The risk–coverage curve records R_k over increasing values of C_k , characterizing the trade-off between answering more questions and maintaining a low error rate [2].

Our primary summary metric is the area under the risk–coverage curve (AURC). Following the repository’s discrete evaluation, AURC is computed as the arithmetic mean of risk over every non-empty ranked prefix:

$$\text{AURC} = \frac{1}{N} \sum_{k=1}^N R_k.$$

Lower AURC indicates a better selective ordering: the system concentrates correct predictions at high-ranking positions and postpones unreliable predictions until higher coverage levels. We additionally report normalized AURC and risk at matched coverage levels as complementary measures.

For interpretability across candidate sets with different full-coverage accuracy, we also report normalized AURC. Let $\text{AURC}_{\text{oracle}}$ denote the AURC obtained by ranking all correct candidates before all incorrect candidates, and let $\text{AURC}_{\text{random}}$ denote the expected AURC under a uniformly random ranking. The normalized metric is

$$\text{nAURC} = \frac{\text{AURC} - \text{AURC}_{\text{oracle}}}{\text{AURC}_{\text{random}} - \text{AURC}_{\text{oracle}}}.$$

Under this normalization, 0 corresponds to the oracle ordering and 1 to the expected random ordering; values above 1 are possible for rankings worse than random. For the binary correctness setting used here, $\text{AURC}_{\text{random}}$ equals the full-set error rate.

Importantly, the final experiment does not select one universal abstention threshold on the held-out test set. Instead, the complete risk–coverage curve evaluates how well each ranking signal supports abstention across all possible coverage levels. A deployment system could subsequently choose an operating coverage or threshold according to its desired trade-off between answer rate and error risk.

4 Experimental Setup

4.1 Dataset and Data Partitioning

We conduct all experiments on SQuAD v2 [7], which contains both answerable and unanswerable questions paired with Wikipedia passages. The presence of unanswerable examples makes the dataset suitable for studying selective question answering and abstention.

The original SQuAD v2 validation split is divided into separate calibration and held-out test partitions using a deterministic 50/50 stratified split with random seed 17. Stratification preserves the proportion of answerable and unanswerable examples across the two partitions.

The calibration partition is used for procedures that require parameter selection before final evaluation. Specifically, it is used to fit the temperature-scaling parameter, inspect calibration behavior, and perform the auxiliary fusion-weight search described below.

4.2 Models

The extractive QA backbone is the pretrained `deepset/roberta-base-squad2` model. The model produces the candidate answers that are shared by all five primary ranking methods.

The question-aware semantic verification path additionally uses `domenicrosati/QA2D-t5-base` to transform question–answer pairs into declarative claims. Valid generated claims are evaluated against the original context using `FacebookAI/roberta-large-mnli`.

The answer-support verifier also uses `FacebookAI/roberta-large-mnli`, but constructs its hypothesis using a fixed question–answer template instead of a generative QA-to-declarative transformation. The two verification paths therefore differ in claim construction but share the same underlying NLI backbone.

None of these pretrained models is fine-tuned as part of the present study. Gold answers and candidate-correctness labels are not supplied to the verification models during inference.

4.3 Calibration Protocol

Temperature scaling is fitted exclusively on the calibration partition. The resulting temperature is

$$T = 4.604539.$$

After fitting, this value is frozen before application to the held-out test set.

The auxiliary fusion-weight analysis is also performed entirely on the calibration partition. For each verifier, the weighted geometric family

$$r_i(\alpha) = c_i^\alpha v_i^{1-\alpha}$$

is evaluated for

$$\alpha \in \{0.00, 0.01, \dots, 1.00\}.$$

Calibration-set AURC is used as the optimization criterion. For both verification signals, the selected value is

$$\alpha = 1.00,$$

corresponding to the confidence-only endpoint. This auxiliary optimization is separate from the primary experiment, in which the two confidence–verification combinations use predefined equal-weight geometric means.

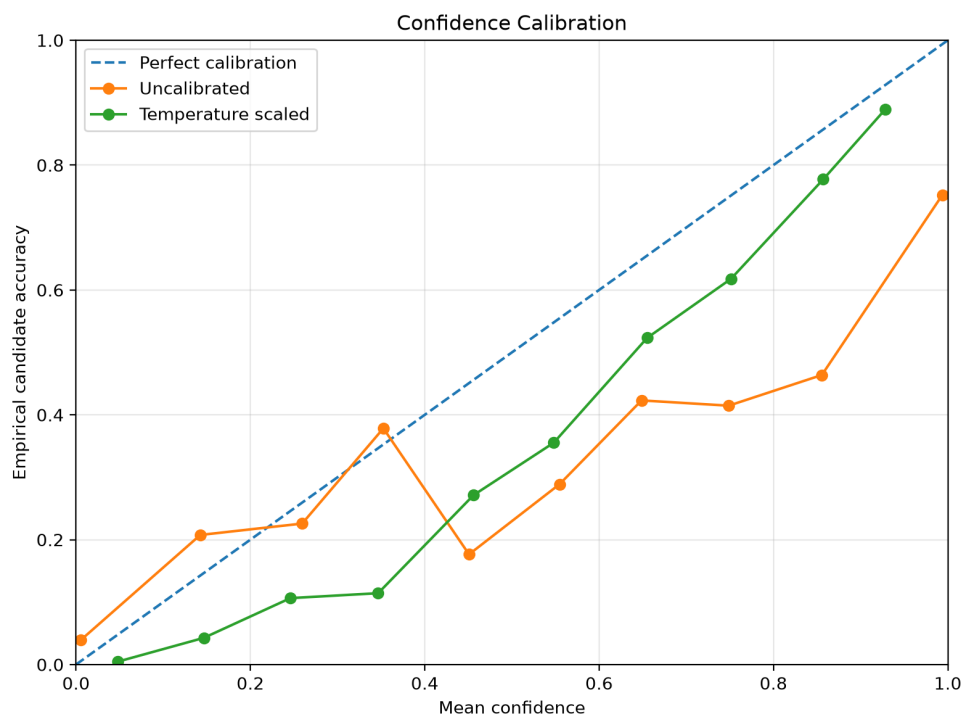


Figure 1: Confidence calibration before and after temperature scaling on the calibration partition. The diagonal represents perfect calibration. Temperature scaling improves the probability scale but, being monotonic, does not change the confidence-only ranking.

4.4 Held-Out Evaluation Protocol

The primary held-out analysis evaluates up to $N = 3000$ examples. A single deterministic shuffled ordering with seed 17 is used to construct the nested evaluation subsets

$$200 \subset 500 \subset 1000 \subset 2000 \subset 3000.$$

Consequently, every larger subset contains all examples from the preceding smaller subset. This design allows changes in method performance with increasing sample size to be inspected without introducing independently sampled evaluation sets.

All five methods are evaluated on exactly the same QA candidates within each subset. Predictions are sorted by ranking score in descending order. When two candidates receive identical scores, the smaller original deterministic evaluation index is used as the secondary ordering key.

The primary comparison consists of the following five ranking methods:

1. calibrated confidence only;
2. question-aware semantic verification only;
3. confidence combined with question-aware verification;
4. answer-support verification only; and
5. confidence combined with answer-support verification.

Because the underlying QA candidates and correctness labels remain fixed, differences among these methods reflect changes in selective ordering rather than differences in answer generation.

4.5 Correctness Definition

The primary experiment evaluates forced-answer candidates. For an answerable example, a candidate is considered correct when its normalized prediction text exactly matches at least one normalized reference answer. For an unanswerable example, a forced-answer candidate is considered incorrect.

This definition is deliberately separated from the model’s native no-answer behavior. A separate native no-answer baseline can count a correct abstention on an unanswerable question as task-level correctness, whereas the primary five-method experiment asks how well different scores rank a common set of forced-answer candidates. The two evaluations therefore represent different operational settings and are not used interchangeably.

4.6 Evaluation Metrics

The primary metric is area under the risk–coverage curve (AURC), with lower values indicating better selective ranking. We additionally report normalized AURC, selective risk and selective accuracy at matched coverage levels, and complete risk–coverage curves.

Evaluation at matched coverage levels provides local operating-point comparisons, whereas AURC summarizes ranking quality over the entire range of possible coverage levels. This distinction allows us to identify cases in which a method performs better at an individual coverage level without producing a better global selective ordering.

4.7 Statistical Uncertainty

For the final $N = 3000$ held-out comparison, statistical uncertainty is estimated using a paired bootstrap procedure with 5000 resamples. The bootstrap seed and deterministic evaluation-order seed are both 17.

For each alternative method m , we compute the paired difference

$$\Delta \text{AURC}_m = \text{AURC}_m - \text{AURC}_{\text{Conf}}.$$

Because lower AURC is preferable, a positive value of ΔAURC indicates an advantage for the confidence-only baseline. We report 95% percentile confidence intervals for these paired differences.

Bootstrap analysis is conducted only after the predictions and ranking rules have been fixed. It is used to quantify uncertainty in the final comparison and is not used for model selection or parameter tuning.

4.8 Leakage Control and Reproducibility

The held-out test labels are not used to fit the temperature parameter, select fusion weights, modify verifier rules, revise claim-validation criteria, choose ranking formulas, or configure the bootstrap procedure. These decisions are fixed before final held-out evaluation.

The principal deterministic operations use seed 17, including dataset partitioning, held-out evaluation ordering, nested-subset construction, and bootstrap sampling where applicable. The reference implementation uses Python 3.12.13 and records exact package versions for reproduction.

The experimental scope is intentionally narrow. The conclusions apply to SQuAD v2, the selected extractive QA backbone, the specified pretrained verification components, and the ranking protocol evaluated here; they are not intended as universal claims about abstention behavior across other datasets, model families, or deployment settings.

5 Results

5.1 Main Held-Out Comparison

We first compare the five ranking methods on the complete held-out evaluation set of $N = 3000$ examples. Because all methods rank the same forced-answer candidates, they share the same full-coverage correctness: 1,267 predictions are correct and 1,733 are incorrect, corresponding to an accuracy of 0.422333.

Table 1 reports AURC and normalized AURC for the five ranking methods. Calibrated confidence achieves the lowest AURC,

$$\text{AURC}_{\text{Conf}} = 0.292379,$$

and therefore provides the strongest global selective ordering in the final evaluation.

The two verifier-only rankings perform substantially worse than confidence. Question-aware verification increases AURC from 0.292379 to 0.394644, while answer-support verification increases it to 0.433892.

Combining verification with confidence improves substantially over using either verifier alone, but neither fixed fusion surpasses confidence. Confidence plus answer-support verification obtains an AURC of 0.309120, while confidence plus question-aware verification obtains 0.339417.

These results indicate that the verification signals contain useful information but do not improve the complete selective ordering produced by calibrated confidence in the final held-out experiment.

Table 1: Final held-out selective-ranking results at $N = 3000$. Lower values are better for both metrics.

Method	AURC	Normalized AURC
Confidence only	0.292379	0.216109
Confidence + answer-support	0.309120	0.262110
Confidence + question-aware	0.339417	0.345358
Question-aware verification	0.394644	0.497105
Answer-support verification	0.433892	0.604947

Table 2: AURC across deterministic nested held-out subsets. Lower is better. Bold values indicate the best method at each sample size.

Method	$N = 200$	$N = 500$	$N = 1000$	$N = 2000$	$N = 3000$
Confidence	0.292532	0.276535	0.276122	0.282511	0.292379
Confidence + answer-support	0.284548	0.278191	0.286815	0.301939	0.309120
Confidence + question-aware	0.349813	0.314886	0.315459	0.330510	0.339417
Question-aware verification	0.419063	0.372894	0.374631	0.388385	0.394644
Answer-support verification	0.425353	0.410249	0.427841	0.434739	0.433892

5.2 Performance Across Sample Sizes

To examine whether the result depends strongly on evaluation-set size, we repeat the same comparison over the deterministic nested subsets of 200, 500, 1,000, 2,000, and 3,000 examples.

Table 2 reports AURC for each method. Confidence achieves the lowest AURC at $N = 500$, 1000, 2000, and 3000. The only exception occurs at $N = 200$, where confidence plus answer-support verification achieves an AURC of 0.284548, slightly below the confidence-only value of 0.292532.

The corresponding full-coverage accuracies of the nested candidate sets are 0.4650, 0.4520, 0.4370, 0.4285, and 0.422333, respectively. These accuracy changes reflect the composition of the nested subsets rather than differences among ranking methods, since all methods operate on the same candidates at each sample size.

The $N = 200$ result shows that verification can produce a local improvement in a smaller sample. However, the ordering changes as the evaluation set grows, and confidence becomes the best-performing method from $N = 500$ onward. The final conclusion is therefore based primarily on the complete $N = 3000$ held-out evaluation rather than the smallest subset.

5.3 Matched-Coverage Behavior

AURC summarizes performance across the complete coverage range, but individual operating points can show a different ordering. We therefore also compare methods at matched coverage levels.

One notable local reversal occurs near 20% coverage in the $N = 3000$ evaluation. At this operating point, confidence plus question-aware verification achieves a selective risk of approximately

$$0.1450,$$

compared with

$$0.1467$$

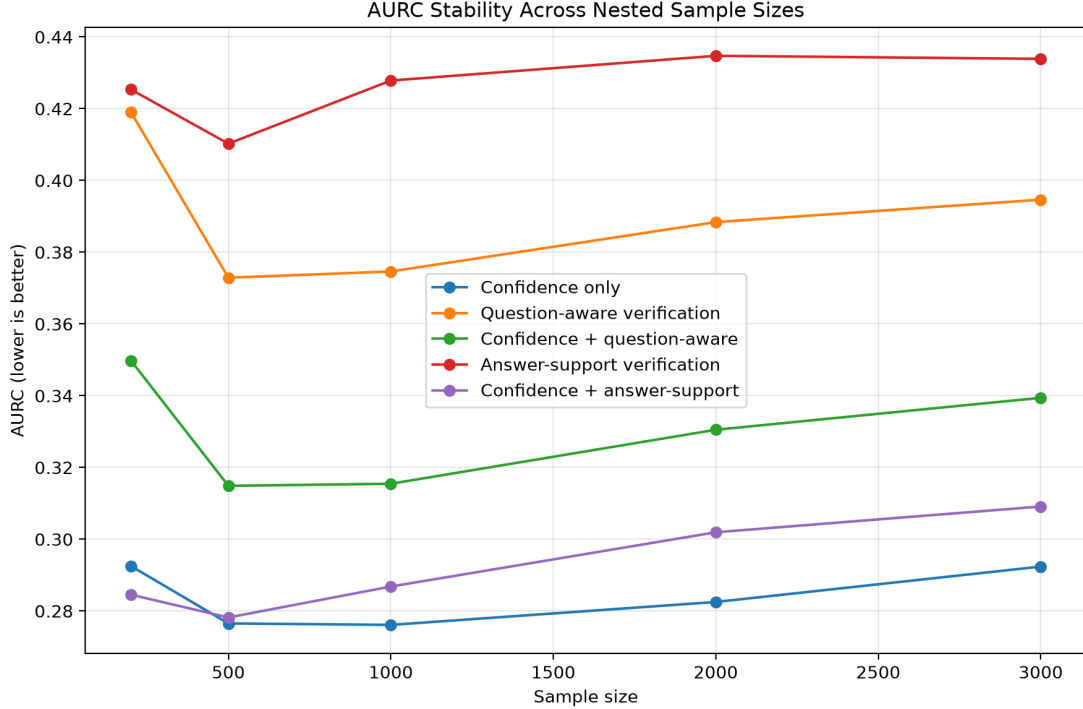


Figure 2: AURC across the deterministic nested held-out subsets. Lower AURC indicates better selective ranking. Confidence is best from $N = 500$ through $N = 3000$, while confidence plus answer-support verification is slightly better at $N = 200$.

for confidence alone.

Thus, at this particular coverage level, the question-aware fusion is slightly better. However, the improvement is local and does not persist across enough of the risk–coverage curve to overcome its poorer global AURC of 0.339417, compared with 0.292379 for confidence.

This distinction is important: a verifier may improve performance at a particular abstention level without improving the overall quality of the ranking across all possible abstention thresholds.

5.4 Paired Bootstrap Analysis

We next assess the stability of the $N = 3000$ comparison using 5,000 paired bootstrap resamples. For each alternative method m , we evaluate

$$\Delta \text{AURC}_m = \text{AURC}_m - \text{AURC}_{\text{Conf}}.$$

Because lower AURC is preferable, positive values favor the confidence-only baseline.

Table 3 reports the observed differences and 95% percentile bootstrap confidence intervals.

All four confidence intervals remain above zero. Under the paired resampling procedure used here, the confidence-only ranking therefore retains an AURC advantage over every alternative throughout the corresponding 95% intervals.

This result should be interpreted as evidence that the observed ordering is stable with respect to resampling of this held-out dataset. It is not intended as a universal statistical claim about other QA datasets, model architectures, or verification systems.

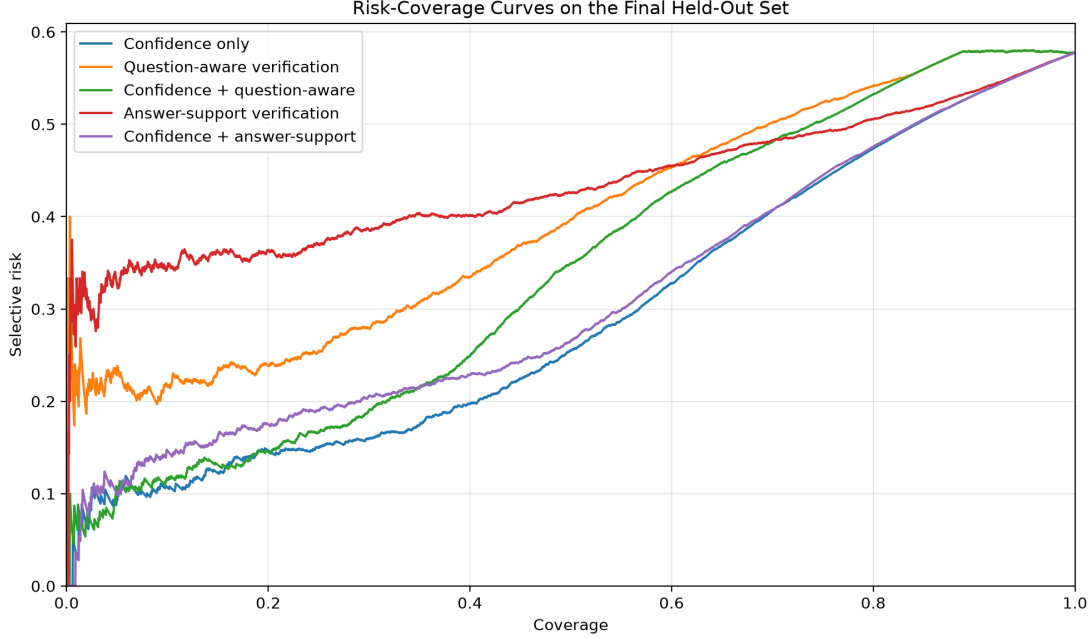


Figure 3: Risk-coverage curves for the five ranking methods on the final $N = 3000$ held-out evaluation. Lower risk is better at a given coverage. Confidence provides the strongest global ordering, although individual coverage levels can exhibit local reversals.

Table 3: Paired bootstrap comparison against confidence-only ranking at $N = 3000$. Positive ΔAURC values favor confidence because lower AURC is better.

Method	ΔAURC	95% CI
Confidence + answer-support	+0.016741	[0.008561, 0.024773]
Confidence + question-aware	+0.047038	[0.036312, 0.057263]
Question-aware verification	+0.102265	[0.086357, 0.118286]
Answer-support verification	+0.141513	[0.124280, 0.159087]

5.5 Calibration-Only Fusion Search

The auxiliary calibration-only fusion experiment provides an additional perspective on the same result. For both verification paths, optimizing

$$c_i^\alpha v_i^{1-\alpha}$$

over

$$\alpha = 0.00, 0.01, \dots, 1.00$$

selects

$$\alpha = 1.00.$$

This endpoint removes the verifier contribution entirely and reduces the ranking to calibrated confidence.

The auxiliary search is not part of the primary held-out comparison, whose fusion rules were fixed in advance as equal-weight geometric means. Nevertheless, its result is consistent with the held-out findings: under the evaluated scoring functions, adding either verification signal does not improve the ranking objective sufficiently to justify assigning it positive fusion weight.

5.6 Summary of Findings

The empirical results yield three main observations. First, calibrated confidence is a strong selective-QA ranking baseline and achieves the best global AURC on the final $N = 3000$ evaluation. Second, combining confidence with verification is consistently better than ranking by either verifier alone, but the fixed combinations still perform worse than confidence alone globally. Third, verification can produce local improvements, such as the $N = 200$ result and the approximately 20%-coverage reversal, without improving the complete risk-coverage ordering.

The remaining question is therefore not whether the verification models produce meaningful semantic signals, but why those signals fail to improve—and sometimes degrade—the ordering of correct and incorrect QA candidates. We examine this question through verifier-level error and ranking analysis in the following section.

6 Error and Ranking Analysis

The results in Section 5 show that both NLI-based verification paths produce worse global selective rankings than calibrated confidence, either alone or under the predefined equal-weight fusion rules. This section examines why. We distinguish two questions: whether verifier outputs are diagnostically associated with QA correctness, and whether they improve the relative ordering of correct and incorrect predictions. These properties are related but not equivalent.

6.1 Question-Aware Verification Labels

The question-aware verifier assigns each valid generated claim an NLI label of entailment, contradiction, or neutral; structurally invalid generated claims are assigned the separate `INVALID_CLAIM` label.

Table 4 shows the relationship between these labels and QA correctness on the final $N = 3000$ held-out set.

Table 4: QA correctness conditioned on question-aware verification labels.

Label	Total	Correct	Incorrect	Accuracy
ENTAILMENT	1622	936	686	0.5771
CONTRADICTION	713	54	659	0.0757
NEUTRAL	332	133	199	0.4006
INVALID_CLAIM	333	144	189	0.4324

The labels are clearly informative at an aggregate level. Predictions assigned `ENTAILMENT` are correct considerably more often than predictions assigned `CONTRADICTION`. In particular, only 54 of the 713 contradiction-labeled candidates are correct.

However, entailment is far from equivalent to QA correctness. Of the 1,622 predictions labeled `ENTAILMENT`, 686 are still incorrect. Thus, semantic support according to the NLI model does not provide a correctness oracle for the extractive QA candidate.

The overlap remains substantial even when considering the continuous entailment score rather than only the categorical label. There are 464 incorrect predictions whose question-aware entailment probability is at least 0.8. Conversely, 273 correct predictions receive an entailment probability of at most 0.2.

These cases illustrate the central difficulty for selective ranking: a verification score can correlate with correctness on average while still assigning high scores to many wrong predictions and low scores to many correct ones.

6.2 Generated-Claim Validity

The question-aware pipeline marks 333 of the 3,000 generated declarative claims as structurally invalid. These claims are assigned a ranking score of zero without running NLI inference.

Among the invalid-claim cases, 144 underlying QA predictions are correct and 189 are incorrect. Their candidate accuracy is therefore approximately

$$0.4324,$$

compared with approximately

$$0.4211$$

among the 2,667 examples with valid generated claims.

This does not imply that invalid claim generation is desirable. Rather, it shows that failure of the QA-to-declarative transformation is not equivalent to failure of the original QA candidate. Assigning every invalid claim a semantic score of zero can therefore strongly down-rank some candidates that are actually correct.

The most frequent validation reasons are shown in Table 5.

Table 5: Structural validation reasons among question-aware generated claims. Reasons are not mutually exclusive.

Validation reason	Count
ANSWER_NOT_PRESERVED	309
NUMBER_NOT_PRESERVED	106
QUESTION_FORM	18
NEGATION_NOT_PRESERVED	5
ANSWER_ONLY_FRAGMENT	3
TOO_SHORT	3

The dominant failure mode is failure to preserve the predicted answer during QA-to-declarative transformation. Numerical preservation is another frequent issue. Because multiple validation rules can be triggered by the same generated claim, the counts in Table 5 do not sum to 333.

This transformation stage introduces a source of ranking noise that does not exist in the confidence-only baseline: the quality of the final score depends not only on the QA prediction and the NLI model, but also on whether the intermediate declarative claim faithfully preserves the information contained in the question-answer pair.

6.3 Answer-Support Verification Labels

The answer-support verifier avoids generative QA-to-declarative conversion by using a fixed hypothesis template. Its diagnostic labels are **SUPPORTED**, **UNCERTAIN**, and **REJECTED**.

Table 6 shows their relationship with QA correctness.

Table 6: QA correctness conditioned on answer-support verification labels.

Label	Total	Correct	Incorrect	Accuracy
SUPPORTED	886	542	344	0.6117
UNCERTAIN	1791	711	1080	0.3970
REJECTED	323	14	309	0.0433

Again, the verifier is diagnostically meaningful. A prediction labeled **SUPPORTED** is substantially more likely to be correct than one labeled **REJECTED**. Only 14 of the 323 rejected predictions are correct.

Nevertheless, the classes still overlap strongly with QA correctness. Among the 886 supported predictions, 344 are incorrect. At the same time, 711 correct predictions fall into the large **UNCERTAIN** group.

The answer-support verifier therefore separates some obvious failures but does not produce a sufficiently clean ordering over the full candidate set to outperform confidence alone.

6.4 Diagnostic Information Versus Ranking Quality

The preceding label statistics might appear inconsistent with the AURC results. Both verifiers distinguish groups with very different empirical accuracies, yet verifier-only rankings perform substantially worse than confidence.

There is no contradiction. Selective prediction depends on the ordering of individual candidates, not merely on average accuracy within broad verifier categories. A useful verifier can identify a subset of clearly poor predictions while still misorder many other correct–incorrect pairs.

This distinction is particularly important for score fusion. The fixed geometric mean changes a candidate’s ranking whenever its verifier score differs from the pattern implied by confidence. Such a change is beneficial only when it moves a correct candidate above an incorrect candidate that confidence had previously ordered incorrectly, or preserves an already useful ordering. If it instead moves an incorrect candidate upward or a correct candidate downward, selective ranking can deteriorate even though the verifier signal remains semantically interpretable.

6.5 Pairwise Ranking Diagnostics

To examine this effect directly, we perform a tie-aware pairwise analysis over every pair containing one correct and one incorrect QA candidate. The final candidate set contains

$$1267 \times 1733 = 2,195,711$$

such pairs.

For each pair, we compare the confidence ordering with the ordering produced by a fixed fusion of confidence and verification. A *strict fix* occurs when confidence ranks the incorrect candidate above the correct candidate and the fused score reverses this error. A *strict harm* occurs when

confidence ranks the correct candidate above the incorrect candidate but the fused score reverses this favorable ordering.

Equal verifier-fusion scores are treated as ties in this diagnostic rather than being classified according to the deterministic index used for AURC computation.

Table 7 summarizes the results.

Table 7: Tie-aware correct–incorrect pair diagnostics for the two fixed confidence–verification fusions.

Fusion	Strict fixes	Strict harms	Net fixes–harms
Confidence + question-aware	75,651	319,327	-243,676
Confidence + answer-support	58,960	105,093	-46,133

For the question-aware fusion, harmful pairwise changes outnumber beneficial changes by more than four to one. The fusion produces 75,651 strict fixes but 319,327 strict harms, giving a net balance of

$$-243,676.$$

It additionally converts 24,100 previously favorable confidence orderings into ties, while only 3,116 unfavorable confidence orderings become ties.

This large imbalance is consistent with the increase in AURC from

$$0.292379$$

for confidence to

$$0.339417$$

for confidence plus question-aware verification.

The answer-support fusion causes fewer changes and therefore remains closer to the confidence baseline. It produces 58,960 strict fixes and 105,093 strict harms, for a net balance of

$$-46,133.$$

Its corresponding AURC increase is smaller:

$$0.292379 \rightarrow 0.309120.$$

Thus, both verifier combinations repair some confidence mistakes, but they damage a larger number of already-correct pairwise orderings. The magnitude of this imbalance also helps explain why the answer-support fusion performs considerably better than the question-aware fusion while still failing to surpass confidence.

6.6 Relation to Calibration-Only Fusion Tuning

The pairwise findings are consistent with the auxiliary calibration-only fusion search. For both verification paths, optimization over the weighted geometric family selects

$$\alpha = 1.00,$$

which is exactly the confidence-only endpoint.

At this value, the verifier receives zero effective fusion weight, so the tuned score produces no verifier-induced ranking changes relative to confidence. The calibration partition therefore independently indicates that the additional verification scores do not provide enough complementary ranking information to justify moving away from the confidence ordering under the evaluated fusion family.

6.7 Why Verification Does Not Improve Global AURC

Taken together, the analyses suggest three mechanisms behind the negative global result.

First, NLI support and QA correctness are related but distinct quantities. A context can appear to support a generated or templated statement even when the extracted answer fails the project’s exact-match correctness criterion. Conversely, a correct QA prediction can receive weak NLI support.

Second, the question-aware path introduces an additional transformation stage. Errors in QA-to-declarative generation, particularly answer and number preservation failures, can assign extremely low verification scores to otherwise correct QA candidates.

Third, fixed fusion changes an already strong confidence ordering. Although verification repairs some confidence mistakes, the pairwise analysis shows that the evaluated verifier scores introduce more harmful than beneficial reordering overall.

The negative AURC result should therefore not be interpreted as evidence that NLI verification contains no useful information. The categorical label statistics demonstrate substantial diagnostic separation, and the results section identifies local operating points where verification can improve selective risk. The narrower conclusion is that, under the present models, score definitions, and fusion rules, this information is not aligned well enough with candidate correctness to improve the complete selective ordering over calibrated confidence.

7 Discussion

The central finding of this study is that additional NLI-based verification does not necessarily improve selective question-answering performance over a strong confidence ranking. On the final $N = 3000$ held-out evaluation, calibrated confidence achieves the lowest global AURC, while both verifier-only methods and both predefined confidence–verification fusions perform worse.

This result is informative because the verification signals are not meaningless. As shown in Section 6, their categorical outputs are strongly associated with QA correctness. Predictions labeled as entailed or supported are considerably more accurate than those labeled as contradicted or rejected. Nevertheless, selective prediction requires more than coarse diagnostic separation. A ranking method must order individual correct and incorrect predictions sufficiently well across the entire candidate set. The pairwise analysis shows that the evaluated verifier fusions repair some confidence-ranking mistakes but disrupt a larger number of already favorable orderings.

7.1 Verification Quality and Ranking Quality

The distinction between verification quality and ranking quality is central to interpreting the results. An NLI verifier addresses whether a textual hypothesis appears supported by the supplied context. The selective-QA objective, by contrast, asks whether a candidate answer should be trusted relative to other candidate answers.

These objectives overlap but are not identical. A hypothesis can receive strong entailment support even when the extracted answer does not satisfy the task’s exact-match correctness criterion. Likewise, an otherwise correct QA candidate can receive weak verification support because of claim construction, model uncertainty, or semantic mismatch between the NLI task and the downstream correctness definition.

Consequently, good semantic verification does not automatically imply good abstention ranking. The verifier must contribute information that is complementary to confidence and aligned with the ordering required by the final selective objective.

7.2 Strength of the Confidence Baseline

The results also emphasize the importance of using a strong confidence baseline. Confidence already contains information about which predictions the QA model considers reliable. In the present experiment, this ordering is difficult for the additional verification signals to improve.

Temperature scaling improves the probabilistic interpretation of the confidence values, but because the transformation is monotonic, it does not change the confidence-only ranking itself. The strong AURC performance therefore reflects the usefulness of the underlying QA confidence ordering rather than a ranking improvement caused by temperature scaling.

This distinction matters when evaluating more complex abstention mechanisms. A verification component may appear useful when compared with an always-answer system or a weak uncertainty heuristic, yet fail to improve upon a well-defined confidence ranking under matched evaluation conditions.

7.3 Why Fusion Can Hurt

The fixed geometric-mean fusion was designed to require agreement between confidence and verification. In principle, this can reduce the ranking of predictions that are confident but poorly supported by the context. The same mechanism, however, can penalize correct high-confidence predictions whenever the verifier assigns them a low score.

This effect is particularly visible in the question-aware verification path. A generated claim can fail structural validation even when the underlying QA candidate is correct, causing its verification score to become zero. More generally, high-confidence predictions can be substantially down-ranked when the verifier disagrees with the QA model.

The pairwise analysis quantifies the consequence. For both verifier fusions, the number of harmful correct–incorrect pair reversals exceeds the number of beneficial repairs. The question-aware fusion introduces the largest net degradation, consistent with its larger increase in AURC. The answer-support fusion preserves more of the original confidence ordering and therefore performs closer to the baseline.

The auxiliary calibration-only fusion search leads to the same qualitative conclusion. For both verification paths, the optimal value within the evaluated weighted geometric family is $\alpha = 1.00$, corresponding to confidence alone. This suggests that the negative held-out result is not solely an artifact of the equal-weight choice.

7.4 Global Performance Versus Local Operating Points

The global AURC result should not be interpreted as evidence that verification is never useful for abstention. The experiments contain several local reversals. Confidence plus answer-support verification is slightly better on the smallest $N = 200$ subset, and confidence plus question-aware verification achieves slightly lower risk near 20% coverage in the final $N = 3000$ evaluation.

These observations illustrate an important distinction between global ranking quality and performance at a specific deployment operating point. A system deployed under a fixed coverage or risk constraint may benefit from a signal that improves behavior in that particular region even if its overall AURC is worse.

Our primary research question concerns the complete selective ordering, so AURC remains the principal evaluation metric. However, future abstention systems may reasonably optimize directly for a deployment-specific operating region rather than for global AURC.

7.5 Implications for Selective QA

The findings suggest that additional verification complexity should be evaluated at the level of the final selective decision rather than only through verifier accuracy or intuitive semantic behavior. A verifier can correctly identify many unsupported answers and still fail to improve the ordering that determines which predictions are answered first.

For future selective-QA systems, verification may be more effective when it is integrated through a learned ranking objective, calibrated specifically against QA correctness, or used conditionally on subsets where confidence is known to be unreliable. Another possibility is to treat confidence and verification as complementary features in a model explicitly trained to predict selective risk rather than combining them with a fixed analytic formula.

These possibilities are hypotheses for future investigation rather than conclusions established by the present experiments. The current evidence supports only the narrower result that the evaluated pretrained NLI signals and fusion rules do not improve global selective ranking over confidence in this controlled SQuAD v2 setting.

7.6 Interpretation of the Negative Result

Negative results are useful when they identify where a plausible method fails under controlled evaluation. The original motivation for this study was that semantic support could provide information unavailable from the QA model’s internal confidence. The experiments confirm that the verifier signals are diagnostically informative, but they also show that this information is not sufficiently aligned with candidate correctness to improve the final ranking.

The resulting conclusion is therefore not that verification is intrinsically ineffective. Rather, additional semantic evidence must be judged by whether it improves the downstream abstention objective. In this setting, increased verification complexity does not translate into improved global risk-coverage performance.

8 Limitations

This study is deliberately controlled in scope, and its conclusions should be interpreted within the boundaries of the evaluated setting.

First, the experiments are conducted on a single benchmark, SQuAD v2, using a single pre-trained extractive QA backbone, `deepset/roberta-base-squad2`. The held-out evaluation is therefore an in-domain assessment rather than a test of robustness under domain shift. Confidence and verification signals may behave differently for other datasets, domains, model architectures, or generative QA systems. In particular, the results do not establish that NLI-based verification is generally ineffective for abstention; they show only that the verification strategies evaluated here do not improve global selective ranking in this experimental setting.

Second, the two verification paths should not be interpreted as independent sources of evidence. Although they construct hypotheses differently, both ultimately rely on the same pretrained `FacebookAI/roberta-large-mnli` backbone. Their errors may therefore be correlated. A comparison involving independently trained verifier architectures, different NLI objectives, or models specialized for answer verification could lead to different conclusions.

Third, the question-aware verification pipeline introduces an additional transformation stage through `domenicrosati/QA2D-t5-base`. Errors in converting a question-answer pair into a declarative claim can affect the verification score independently of the quality of the original QA prediction. This limitation is visible empirically: 333 generated claims are marked structurally invalid in the final evaluation, including 144 cases in which the underlying QA prediction is correct. Because invalid claims receive a semantic score of zero, failures of the intermediate representation can directly induce harmful down-ranking.

Fourth, the primary score combinations use predefined equal-weight geometric means rather than a learned fusion model. The auxiliary calibration-only search partially addresses this concern by evaluating a weighted geometric family and selecting $\alpha = 1.00$ for both verification paths, corresponding to confidence alone. Nevertheless, this search explores only a restricted one-parameter family. More expressive approaches could model nonlinear interactions between confidence and verification, condition verifier use on particular candidate types, or optimize directly for selective risk.

Fifth, the main comparison evaluates rankings over fixed forced-answer candidates. For answerable questions, correctness is defined by normalized exact match with a reference answer; for unanswerable questions, a forced-answer candidate is treated as incorrect. This design is useful because it isolates the quality of the ranking signal, but it is not identical to end-to-end QA deployment, where a model may explicitly return no answer. The separate native no-answer setting therefore represents a different operational objective and should not be conflated with the five-method ranking comparison.

Sixth, normalized exact match provides a reproducible correctness criterion but is stricter than semantic equivalence. A prediction can be meaningfully compatible with the reference while failing exact match, whereas an NLI verifier evaluates a different semantic relation between hypothesis and context. Part of the observed disagreement between verification scores and QA correctness may therefore arise from a mismatch between the verifier’s semantic objective and the downstream evaluation criterion rather than from verifier failure alone.

Seventh, statistical uncertainty is quantified with paired bootstrap resampling of the final $N = 3000$ held-out set. These intervals characterize the stability of the observed method differences under resampling of this evaluation sample. They do not capture variation across independently drawn datasets, alternative pretrained model checkpoints, different QA backbones, or repeated end-to-end experimental pipelines. The bootstrap results should therefore be interpreted as conditional evidence for the present held-out comparison rather than as a universal significance statement.

Finally, the primary evaluation summarizes selective performance using global AURC. This is appropriate for comparing ranking quality across the complete coverage range, but it may obscure improvements that occur within a narrow deployment-relevant operating region. The observed local reversals—including the advantage of confidence plus question-aware verification near 20% coverage—illustrate this distinction. A system designed for a specific coverage or risk constraint could reasonably prefer a method that is not globally optimal under AURC.

Taken together, these limitations define the intended scope of the result. The study does not claim that semantic verification cannot improve abstention in general. Rather, it provides controlled evidence that, for the evaluated SQuAD v2 extractive-QA system, the tested pretrained NLI signals and fusion strategies do not improve the complete selective ordering produced by the confidence

baseline. Broader conclusions require evaluation across additional datasets, model families, verifier architectures, and deployment objectives.

9 Conclusion

This study examined whether NLI-based semantic verification can improve selective question answering beyond the ranking information already provided by model confidence. Under a controlled SQuAD v2 protocol, five ranking methods were compared over the same forced-answer candidates, allowing differences in risk-coverage performance to be attributed directly to the ranking signal rather than to changes in answer generation.

The central empirical result is that the confidence baseline provides the strongest global selective ordering in the final held-out evaluation. At $N = 3000$, confidence achieves an AURC of 0.292379, compared with 0.309120 for confidence combined with answer-support verification and 0.339417 for confidence combined with question-aware semantic verification. Both verifier-only rankings perform worse. Paired bootstrap analysis further shows that the AURC differences between confidence and all four alternatives remain positive across the corresponding 95% bootstrap intervals on the evaluated held-out sample.

The error and ranking analyses clarify why this result does not imply that the verification signals are uninformative. Both verifiers distinguish groups with substantially different empirical correctness rates, and verification occasionally improves performance at specific sample sizes or coverage levels. However, selective prediction depends on the ordering of individual candidates rather than on verifier plausibility alone. Under the fixed fusion rules evaluated here, verifier-induced changes repair some confidence-ranking errors but disrupt a larger number of already favorable correct-incorrect orderings. The question-aware path is additionally affected by failures in the intermediate QA-to-declarative transformation, which can assign very low verification scores to otherwise correct QA predictions.

The resulting conclusion is therefore deliberately narrower than the claim that semantic verification is ineffective. In this experimental setting, the tested pretrained NLI signals do not provide sufficiently complementary ranking information to improve global AURC over the confidence baseline. The findings instead highlight a broader methodological point: additional verification should be judged by its effect on the downstream abstention objective, not merely by whether the verifier appears semantically meaningful in isolation.

Knowing when not to answer is therefore not simply a matter of adding another layer of verification. A useful abstention mechanism must improve the ordering of reliable and unreliable predictions in the region that matters for the intended decision policy. Future work should test whether this can be achieved through learned fusion, verifier models optimized directly for QA correctness, broader cross-domain evaluation, or objectives targeted at deployment-specific coverage and risk constraints.

Generative AI Disclosure

Generative AI tools, primarily OpenAI’s ChatGPT, were used extensively throughout this project. Their use included assistance with code generation and refactoring, debugging, development of experimental and evaluation workflows, analysis scripting, figure-generation code, interpretation and organization of experimental results, and substantial drafting and language revision of the manuscript. Substantial portions of both the implementation and manuscript prose were produced with generative-AI assistance.

The author executed the experiments, inspected the resulting artifacts and outputs, reviewed and revised the generated code and text, and approved the final experimental protocol, analyses, interpretations, and manuscript. The author retains full responsibility for the accuracy of the reported results, the scientific claims, and the final content of the work. Generative AI systems are not listed as authors and are not treated as independent sources of empirical evidence.

Code Availability

The source code, experiment configurations, evaluation scripts, and version-controlled analysis artifacts supporting this study are publicly available in the project’s GitHub repository. The results reported in this manuscript correspond to repository commit `a871b08`. This snapshot also contains the publication-facing figure-generation script used to regenerate the reported evaluation plots from the canonical tracked artifacts.

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